Modifications To One Shape BO Package

14 October 2026

1 Introduction

This follows the report descriptively named '`261007 Additional Notes`' (see '`261012_One_Shape_BO_Package_Info_Notes.pd`
at `C:\Users\pemb6626\OneDrive-Nexus365\Documents\POSTDOC\Code\Topology_Optimisation\Bayesian_Optimisation_`
`Part3`).

1.1 Problems

Tests were run, summarised in `261009_One_Shape_BO_Runs` (same location as above). The following questions were asked:

1. Why is the result in `Triangle_1` so different to that in `Triangle_0`, despite only changing the seed (in addition to the respective weights)? I am not confident that the code is finding the near-global optimum, as the result can vary greatly between different seeds.
2. Can it be concluded that changing the acquisition/novelty weighting has negligible difference on the final compliance? Why/not?
3. There is no obvious trend in behaviour when the proportion of local and global seeds are varied (`Triangle_2`). Is this conclusive, or should further checks be made (e.g. increasing the total `ng_rst` + `nl_rst`) and why/not?
4. The ellipse outputs are visually very different. Arguably more than the variation in triangle results. Is there a mathematical or computational reason behind thi, or is this a consequence of only varied seeds?

1.2 Analysis

These runs do not establish that the optimiser reliably finds a near-global optimum, that acquisition/novelty weighting is negligible, or that any local/global restart ratio is preferable!

Two concrete computational issues are visible:

1. The seed-76 triangle runs suffer a severe GP-model collapse
2. The ellipse GP representation does not respect exact geometric symmetries, so identical ellipses can appear far apart to both the GP and novelty calculation.

1.3 Geometry Audit

Table 1: Comparison of geometry for each dataset in the Tests folder

Dataset	Treatments and final compliance	Important behaviour
`Triangle_0`	40/60: 80.03; 20/80: 79.58; 70/30: 80.27	32-36 improvements; final improvements at iterations 262-479
`Triangle_1`	All three weights: 106.43	One improvement at iteration 6, then 519 iterations without improvement
`Triangle_2`	5G/10L:82.62; 8G/7L: 92.81; 12G/3L: 93.01	Nominal ratios were partly overridden by stagnation control
Ellipse	Seed 976: 122.46; Seed 32: 138.69; Seed 83: 138.718	Last improvements at iterations 459, 402 and 487 respectively

Visually:

- `Triangle_0` consistently finds a Warren-truss-like family: a sloping left triangle, upper-middle opening, and lower-right opening, leaving strong diagonal material paths.
- `Triangle_1` images are byte-identical and remain in a bulkier, less-efficient basin.
- `Triangle_2` reaches three noticeably different geometric basins.
- `Ellipse` seed 976 has three comparatively balanced holes. Seeds 32 and 83 produce a very large left ellipse, a small lower-middle ellipse, and an upper-right ellipse. Seed 83 exaggerates the large/small size split.

2 Question 1

Why is Triangle 1 so different from Triangle 0? This is not merely ordinary variation between local optima; seed 76 causes the GP surrogate to become effectively uninformative.

Evidence:

- Seed 76 begins with a better burn-in incumbent: 09.15 versus 125.93 for seed 967. Therefore, a poor initial incumbent is not the explanation.
- At BO iteration 11, all seed-76 runs evaluate a compliance of approx. 1979.46
- On the following GP fit, intferred likelihood noise jumps from about 0.008 to 0.689. exceeds 0.91 by iteration 27, and ends near 0.983.
- At the same time, mean output scale falls to approximately 0.000894 and posterior standard deviation to roughly 0.0012.
- The GP has therefore explained almost all observed variation as noise, while retaining almost no latent signal. LogEI then provides little useful direction.
- Seed 967 does not show this failure: noise reaches its 10^{-4} lower bound and its output scales remain approximately 0.28-0.45.

The FEM response is deterministic, but rasterisation makes it discontinuous: changing a ligament by one cell can create a very large compliance jump. The current model instead learns homoscedastic observation noise because no fixed `train_Yvar` is supplied in `gaussian.py` (line 49). What does this last line mean?

Homeoscedasticity: the variance of errors or residuals is constant and equal across all levels of an independent variable or predictor in a statistical model.

This also explains why increasing burn-in or restart count is not the immediate answer. Seed 76 already had a good burn-in point, and additional restarts cannot rescue an acquisition function derived from an almost constant surrogate.

A finite 630-evaluation run in an 18-dimensional constrained, rasterised space cannot certify near-globality. Moreover, theoretical EI convergence requires restrictive fixed-prior/RKHS assumptions: standard sequentially estimated GP parameters can fail to find the optimum even asymptotically. Elaboration on this paragraph required.

Proposed computational change:

- Treat the FEM as deterministic by supplying a fixed, small numerical observation variance instead of fitting homeoskedastic noise.

  See `botorch.org/docs/models`: noise can be treated in several different ways. Where is the noise provided (line of code). How can a fixed noise be provided without proper noise estimates (or can the noise be estimated?)

- Choose the smallest positive numerical nugget that remains well conditioned after the negative-log transform: test a short predefined range rather than silently choosing one. What does 'positive numerical nugget' mean? What defines if it is 'well conditioned' or not? How to test a short predefined range?

- Add a health check using output scale, posterior spread, and signal-to-noise ratio. If a warm-started model degenerates, perform one cold refit and record the event. Elaboration on this 'health check' is required. What does a degenerate model look like?.

- Persist every evaluated design vector, compliance, raster fraction, geometry signature and source. The present artifacts are sufficient to diagnose collapse temporarily, but not to replay the exact seed-76 GO fit and prove that the iteration-11 point alone caused it. 'Persist'? What does this mean?.

- Compare the fixed-noise and current models over paired seeds, reporting collapse frequency, final compliance, best-so-far area under the curve, and evaluations-to-target—not just the best endpoint. Anytime/evaluations-to-target reporting follows established black-box benchmarking practice described by COCO (see the docs). I need a clear structure for these tasks for evaluation, e.g. how many seeds should I run?

3 Question 2

Are acquisition/novelty weights negligible?

No. The evidence only supports the narrower statement that, for healthy seed 967 and these three settings, final compliance differed by less than 0.9%. The trajectories were not equivalent:

- They first diverged as early as iteration 7.
- Their last improvements occurred between iterations 262 and 479.
- Their global/local winner counts and optimisation-failure counts differed substantially.

`Triangle_1` supplies no useful replication because every incumbent froze at iteration 6 before the later trajectories diverged.

There is also an implementation reason for the apparently weak effect. The weights in [`rank_initial_condition_pool` (line 580)] apply only to global raw starts. Then [`take_restart_batch` (line 706)] selects:

- The first start principally by weighted rank; then
- Later starts principally by maximum distance from already selected starts.

Weighted rank is only a tie-break after that. Thus, the requested weights primarily affect the first start and duplicate/tie cases, rather than controlling the whole restart batch. Oh! I thought that every global start is selected by weight from already-selected starts. Therefore for every ng rst, the weighting adjusts according to already selected starts. Would my interpretation be a better way to cover the domain?

Proposed computational change:

- Add acquisition and novelty weights to OptimizationConfig, validate and normalize them, and save them in the run configuration and log.

- If weights are intended to control every global start, use a greedy maximum-marginal-relevance score at each selection:

$$S_i(A) = w_a \widetilde{\mathrm{LogEI}}_i + w_n \widetilde{d}(x_i,\ X_{\text{observed}} \cup X_{\text{attempted}} \cup A) \tag{1}$$

Where A is the set already selected and both terms are scaled to $[0, 1]$. This retains LogEI as the BO acquisition function; it only makes restart-initialisation semantics match the advertised weights. Retain a legacy-selection option until paired comparisons show that the new selection is no worse. Does the first part of this agree with my interpretation, as discussed in the previous blue comment? What is meant by this 'legacy-selection' option? Is it only necessary for investigating the code's robustness?

4 Question 3

The study into the proportion of global/local restarts is not conclusive, and increasing the total number of restarts blindly is not the appropriate next step.

Although the three `Triangle_2` runs alone appear to favour more local starts, the 10G/5L baseline reaches 79.9892, outperforming all three. There is therefore no monotonic overall relationship.

Several confounders explain this:

- The stagnation controller changes nominal allocations. The 8G/7L run spends 166 iterations at 12G/3L; the 10G/5L baseline spends 203 iterations at 12G/3L.
- A "local" restart describes its initial source, not its search domain. All starts are subsequently optimized over the full box in [Acquisition (line 358)]. This is consistent with the repository documentation that archive mode changes initial conditions, not LogEI.
- Consequently, a 12G/3L run can still report 451 local winners: those three archive starts can move throughout the domain and repeatedly beat global starts.
- One shared random generator is used for burn-in and later sampling. Changing restart counts changes random-number consumption, so later candidate pools are not nested even with the same initial seed.

For a fixed GP and genuinely nested restart sets S_R,

$$\max_{x \in S_{R+1}} a(x) \geq \max_{x \in S_R} a(x).$$

That only guarantees a non-decreasing optimized acquisition value, not better true compliance. BoTorch likewise describes multiple restarts as an approximation to the maximum of a non-convex acquisition function, not to the physical objective itself. `https://botorch.org/docs/optimization` What exactly are 'nested restart sets'? Does this simply mean ensuring that the seed is allocated per iteration rather than at the start (which is not comparable as changing restart counts change the random number consumption).

Proposed computational change:

- Add an explicit way to disable adaptive allocation during controlled ratio studies; setting patience beyond the run length can provide the same experimental control initially.
- Use phase- and iteration-specific RNG streams, or save one master raw pool, so restart budgets can be compared with genuinely nested start sets. Does this address my previous query in blue?
- At representative frozen GP states, test restart totals such as 5, 10, 15 and 30 using nested starts. Measure best feasible LogEI, unique optimized endpoints and wall time. Increase the production total only if acquisition quality has not saturated.
- Then compare selected totals across paired full-run seeds.
- If "local" is intended to mean a mathematically local search, archive starts must instead have lengthscale-scaled trust-region bounds. A principled scheme expands after successes and contracts after failures, as in TuRBO. That would be a separate modelling change, not a reinterpretation of the existing restart ratio.

Is my interpretation of the proposed computational change correct:

- Change 1: This is purely for fair monitoring of global vs local restarts. This is bullets 1-4.
- Change 2: This is a proposal to 'improve' local search. Currently, the `nl_rst` best archived restarts are used as new restarts. These best archived restarts are ordered by GP kernel distance. Selection is currently biased towards diversity. What might this Change look like?

5 Question 4

Why do ellipse geometries vary so much? Seed-dependent basin selection and objective non-uniqueness both matter, but there is also a definite ellipse-specific representation problem.

The data show two distinct effects:

- Seed 976 versus seeds 32/83 produces both substantially different masks and approximately 13.3% better compliance.
- Seeds 32 and 83 differ over 13.2% of the domain but have virtually identical compliance—only 0.030 apart. Thus, compliance does not uniquely identify one geometry.

Ellipse runs actually have fewer active continuous variables than triangle runs, so this is not explained by higher dimensionality.

- Triangle canonicalisation (line 110) removes vertex-order, orientation and hole-order equivalences.
- Ellipse canonicalisation (line 151) only sorts holes by centre.

Physically,

$$E(a, b, \theta) = E(a, b, \theta + \pi) = E(b, a, \theta + \pi/2),$$

and when $a = b$, orientation is completely irrelevant. Nevertheless, the raw Matérn GP and raw Euclidean novelty metric treat these encodings as different. In particular, normalized angles 0 and 1 produce the same ellipse but lie one full unit apart. This wastes observations and creates artificial acquisition modes and discontinuities at the angle boundary. Periodic input mappings are standard GP constructions; see Rasmussen and Williams, Chapter 4. More generally, known invariances can be incorporated into BO kernels to improve sample efficiency.

Proposed computational change: Represent each physical ellipse by its symmetric-positive-definite shape matrix

$$S = R_\theta \begin{bmatrix} a^2 & 0 \\ 0 & b^2 \end{bmatrix} R_\theta^\top,$$

and use the GP feature vector

$$\phi = (c_x/x_{\text{size}},\ c_y/y_{\text{size}},\ L_{11}, \sqrt{2}L_{12}, L_{22}), \qquad L = \log S.$$

This representation:

- Is identical under $\theta + \pi$;
- Is identical under an axis swap plus $\pi/2$;
- Becomes independent of angle for circles;
- Preserves all geometric information about the ellipse.

Use $k(\phi(x), \phi(x'))$ for the GP and the same transformed-space distance for ellipse novelty/archive diversity. Keep raw variables for acquisition optimization, constraints, rasterisation and FEM, so no physical model changes. As a secondary check, compare the current 100-direction ellipse-clearance approximation against a much finer reference near the spacing boundary. There is not presently enough evidence to blame that approximation, so it should not be changed unless the audit finds material errors. Approximately, which functions will this change affect? Briefly, what might this change look like?

6 Responses to Further Questions

IMPORTANT! See latest additions to the docs folder, particularly the papers. Skim read. Important because they are referenced below and aid understanding, Ellie! DON'T REGURGETATE.

6.1 Question 1: GP Collapse and Reliability

6.1.1 What does the absence of fixed observation noise mean?

A Gaussian Process (GP) commonly models an observation as

$$y_i = f(\mathbf{x}_i) + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma_n^2).$$

Here, f is the underlying response and σ_n^2 is the observation-noise variance. When `train_Yvar` is not supplied, the GP can estimate σ_n^2 from the training data. In the usual homoscedastic model, the same variance applies to every observation. In this problem, the FEM is nominally deterministic: repeating precisely the same geometry, mesh and solver settings should produce the same compliance. However, small changes in geometry can produce large compliance changes when rasterisation creates or removes a narrow ligament. These changes are physical variations in the computed objective, not necessarily observation noise. Consequently, the GP may incorrectly attribute difficult-to-model response variations to noise. A sufficiently large fitted noise variance can suppress the GP's estimated signal, producing an almost constant surrogate.

Important distinction: A deterministic function need not be smooth. Specifying near-zero observation noise does not, by itself, make a smooth GP kernel suitable for discontinuous behaviour.

6.1.2 Why can near-global optimality not be established?

The optimisation seeks

$$C^* = \min_{\mathbf{x} \in \mathcal{F}} C(\mathbf{x}),$$

where $\mathcal{F}$ is the feasible design space. A finite optimisation run only establishes

$$C^* \leq C_{\text{best}},$$

where C_{best} is the lowest compliance observed. It does not establish how close C_{best} is to C^*. This is particularly important here because:

- The design space is high-dimensional, constrained and non-convex.
- Rasterisation introduces sharp response changes.
- Different starting designs can lead to different geometric basins.
- The GP and acquisition optimisation are approximations.

Theoretical convergence results for Expected Improvement generally rely on assumptions about the objective, kernel and GP-prior treatment that are not automatically satisfied here.

For example, Bull (2011) establishes convergence properties under particular fixed-prior assumptions, but also demonstrates that certain procedures which repeatedly estimate GP hyperparameters can fail to locate the optimum asymptotically.

See `https://www.jmlr.org/papers/v12/bull11a.html`. Therefore, the immediate objective should be robust, reproducible optimisation performance across seeds, rather than claiming near-global optimality.

6.1.3 Where should observation noise be specified?

The relevant location is the GP construction in gaussian.py, reported as being around line 49 in the existing audit. For BoTorch's SingleTaskGP, supplying `train_Yvar` fixes the observation variances rather than estimating one common noise variance. These values have the same batch/output dimensions as the training responses.

There are two different quantities to consider:

1. Observation noise: Variation when evaluating exactly the same input repeatedly.

2. Model discrepancy: Differences between the true FEM response and what the GP kernel can represent.

The first can be estimated using repeated identical evaluations, ideally on the same computational setup. The second cannot generally be estimated as observation noise by repeating an identical evaluation. An initial fixed-noise model is therefore reasonable, provided its numerical variance is tested rather than assumed to be physically measured. The noise must also be specified consistently with the modelled response. If compliance is transformed using $y = -\log C$, the relevant variance is that of y, not the original compliance. Any subsequent output standardisation must also be accounted for.

6.1.4 What is a numerical nugget, and how should it be tested?

A numerical nugget is a small positive diagonal regularisation term:

$$K_{\text{reg}} = K + \nu I, \qquad \nu > 0,$$

where K is the training covariance matrix. It improves numerical stability when K is nearly singular, for example when training points are very close. Here, ν is a variance, not a standard deviation. A practical initial test is

$$\nu \in \{10^{-8}, 10^{-6}, 10^{-4}\},$$

expressed in standardised GP-output variance units. These are candidate numerical values, not estimates of physical noise. For each candidate, examine:

- Whether GP fitting completes without numerical failures.
- Whether covariance factorisation is stable, without repeated escalation of numerical jitter.
- Whether fitted hyperparameters remain finite and away from pathological bounds.
- Whether posterior predictions and uncertainties remain informative.
- Whether predictive performance improves on held-out observations.

The selected nugget should be the smallest tested value providing stable fitting and acceptable predictive behaviour. A larger nugget is not automatically better. If the chosen GP implementation internally standardises responses, convert the prescribed standardised variance to the units expected by its observation-noise interface.

6.1.5 What constitutes a degenerate GP?

A GP should be considered potentially degenerate when it ceases to represent meaningful variation in the objective. Useful diagnostics are:

$$\text{SNR} = \frac{\sigma_f^2}{\sigma_n^2}, \qquad \eta = \frac{\sigma_f^2}{\sigma_f^2 + \sigma_n^2},$$

where σ_f^2 represents the GP signal variance and σ_n^2 its observation-noise variance. A collapse may be indicated by the combination of:

- Very small signal variance relative to the observed output variation.
- Very large inferred noise relative to the signal.
- Almost constant posterior mean across representative feasible designs.
- Very small posterior standard deviation across those designs.
- Poor predictive agreement with withheld evaluations.

For example, $\eta < 0.01$ could be investigated as an initial warning threshold, but no individual threshold should automatically declare model failure. When a warning is triggered:

- Record the training data, hyperparameters and diagnostic values.
- Refit once from a fresh hyperparameter initialisation, rather than using the previous warm start.
- Compare the two fits using likelihood, predictive diagnostics and numerical stability.
- Record whether the warning persists.

With fixed observation noise, the noise-based SNR diagnostic becomes less informative; signal variance, posterior behaviour and predictive performance must still be checked. The unusually large seed-76 observation should also be independently re-evaluated and its rasterised geometry inspected. The existing logs suggest a relationship between that observation and GP collapse, but do not prove causation.

6.1.6 What does 'persist' mean?

To persist information means to save it permanently to disk, rather than retaining it only in memory or printing it to the terminal. For each evaluated design, store:

- Iteration number, seed and complete encoded design vector.
- Compliance and transformed GP response.
- Rasterised volume fraction and feasibility information.
- Geometry signature and source of the selected candidate.
- GP hyperparameters, health diagnostics and fitting warnings.
- Acquisition/restart information and relevant random-generator states.

This would allow the exact training dataset preceding iteration 11 of seed 76 to be reconstructed, inspected and used to refit alternative GP models without repeating the complete optimisation.

6.1.7 Recommended experimental structure

A staged investigation is preferable to immediately repeating many complete 630-evaluation optimisations.

Table 2: Proposed GP reliability investigation

Stage	**Experiment**	**Purpose**
A: Diagnosis	Replay stored GP training datasets immediately before and after the seed-76 failure; compare learned noise against several fixed nuggets.	Identify whether the failure is reproducible and sensitive to noise modelling.
B: Pilot	Use 3 paired seeds, including 76 and 967, over approximately 150–200 BO iterations per configuration.	Eliminate unstable GP configurations at moderate computational cost.
C: Confirmation	Compare the current GP and the preferred alternative using 10 predetermined paired seeds and the same full evaluation budget.	Estimate reliability and performance differences across initial conditions.
D: Extension	Increase to approximately 20–30 paired seeds if differences are small or highly variable.	Improve precision before concluding that one method is preferable.

A paired seed means that both methods begin with the same initial design and equivalent controlled randomness. Their subsequent BO evaluations may naturally diverge. The main performance measures should be:

- Final compliance: median, spread and paired differences.

- Frequency of GP-collapse warnings and numerical failures.
- Best-so-far compliance curves.
- Area under the best-so-far compliance curve, with smaller values preferred.
- Number of evaluations required to reach one or more predefined compliance targets.

The compliance targets and evaluation budgets should be selected before comparing methods. Report uncertainty intervals for paired performance differences. This follows the anytime/target-based benchmarking principles of COCO.

6.2 Question 2: Acquisition and Novelty Weighting

6.2.1 Should the weighting be updated for every global start?

Yes. This agrees with the proposed modification and with the intended interpretation. Under the reported current behaviour, acquisition/novelty weighting primarily determines the initial global restart ranking, whereas subsequent selection places greater emphasis on distance. The alternative is to recompute novelty after selecting each new global restart. For candidate $\mathbf{x}_i$ and already-selected global starts A, define

$$S_i(A) = w_a \widetilde{a}(\mathbf{x}_i) + w_n \widetilde{d}\left(\mathbf{x}_i,\, X_{\text{observed}} \cup X_{\text{attempted}} \cup A\right),$$

subject to

$$w_a + w_n = 1, \qquad w_a, w_n \geq 0.$$

At each selection:

1. Score all eligible remaining global candidates.
2. Select the candidate with the largest $S_i(A)$.
3. Add that candidate to A.
4. Recalculate distances and scores for the remaining candidates.
5. Continue until `ng_rst` starts have been selected. Both score components should be consistently scaled to $[0, 1]$.

This is a greedy acquisition/diversity selection rule. It directly implements the intended weighting for every global start, rather than principally the first. It should generally discourage clustered initial conditions, but it does not guarantee better final compliance: distinct initial starts may still converge to the same acquisition optimum. Importantly, LogEI remains the BO acquisition function. The weighted score only chooses where acquisition optimisation begins.

6.2.2 What is a legacy-selection option?

A legacy option simply preserves the existing restart-selection procedure alongside the proposed procedure. For example, an experimental configuration could select between:

- Existing acquisition/novelty ranking followed by distance-prioritised selection.
- New greedy selection that recalculates the weighted score after every selected start.

Its purpose is primarily experimental: it allows controlled comparisons and regression testing without permanently losing the baseline. It is not inherently necessary in the final production implementation once the preferred procedure has been validated.

6.3 Question 3: Global and Local Restart Allocation

6.3.1 What are nested restart sets?

Nested restart sets contain progressively more initial conditions while retaining all previous ones:

$$S_5 \subset S_{10} \subset S_{15} \subset S_{30}.$$

For example, the 10-restart experiment uses exactly the same first five starting points as the 5-restart experiment, plus five additional points. This permits a controlled test of whether additional restarts improve optimisation of the same acquisition function. Assigning a random seed per iteration is not sufficient on its own. It prevents randomness in earlier iterations from changing later random streams, but nested sets additionally require:

- The same frozen GP and acquisition function.
- A shared candidate pool or reproducible ordered candidate sequence.
- Deterministic selection of prefixes of that sequence.
- Equivalent optimisation settings for shared initial conditions.

If the acquisition optimisation reliably returns the best successful restart result, adding restarts cannot reduce the best acquisition value obtained from that fixed set. It can nevertheless fail to improve the actual physical objective. In complete BO runs, different restart budgets select different evaluations, so their future GP models and trajectories diverge. Nested-set reasoning therefore applies directly to frozen-GP experiments, not automatically to full BO trajectories.

6.3.2 Do independent random-number streams solve the comparability problem?

Partially. Independent streams for burn-in generation, global candidate sampling, local candidate selection and individual BO iterations prevent one component's random-number consumption from inadvertently changing another component. However, to compare restart counts using nested sets, candidate generation and restart ordering must also be controlled explicitly. The two measures are complementary: `Independent RNG streams + shared ordered start pools =`

6.3.3 Are there two distinct proposed changes?

Yes. **Change 1: Fair evaluation of the existing approach.** The first four modifications in Question 3 concern experimental control:

- Disable adaptive restart-allocation changes.
- Separate random-number streams.
- Compare nested restart budgets on frozen GP models.
- Confirm promising settings using paired full BO runs.

These changes are intended to establish whether the current method benefits from different restart numbers or ratios. They are not, by themselves, a new local-search algorithm.

Change 2: Introduce genuinely local acquisition optimisation. At present, archive-based local restarts provide historically promising initial conditions. Their diversity may be controlled using the GP kernel distance, but subsequent acquisition optimisation can move outside the neighbourhood of the archived design. A local-search modification would instead:

- Select promising, geometrically distinct archived designs.
- Construct a bounded neighbourhood around each selected design.
- Optimise LogEI within that neighbourhood, respecting the existing physical constraints.

- Expand the neighbourhood after repeated improvements.
- Contract it after repeated unsuccessful searches.
- Retain global restarts to search outside those neighbourhoods.

For an archive point $\mathbf{x}^{(a)}$, a simple local region is

$$\mathcal{T}_a = \left\{ \mathbf{x} : |x_j - x_j^{(a)}| \le r_j \right\} \cap \mathcal{F},$$

where r_j defines the neighbourhood width in each active coordinate. The widths could reflect GP lengthscales where these correspond directly to the optimisation coordinates. If the GP subsequently uses transformed ellipse features, the lengthscales cannot simply be reused as raw-coordinate widths. Adaptive trust regions are related to the ideas behind TuRBO. However, adding bounded archive-centred optimisation to the current global GP would be trust-region-inspired, not a complete implementation of TuRBO. See `https://botorch.org/docs/v0.13.0/tutorials/turbo_1` This should be investigated separately from the restart-ratio study.

6.4 Question 4: Ellipse Representation

6.4.1 Which functions would change?

The proposed ellipse representation changes the features seen by the GP and diversity calculations, not the physical ellipse definition. For ellipse semi-axes a, b and orientation θ:

$$S = R_\theta \begin{bmatrix} a^2 & 0 \\ 0 & b^2 \end{bmatrix} R_\theta^\top.$$

For dimensionless features, define

$$D = \begin{bmatrix} x_{\text{size}} & 0 \\ 0 & y_{\text{size}} \end{bmatrix}, \qquad \widehat{S} = D^{-1} S D^{-1}.$$

The GP feature vector is then

$$\phi(E) = \left(\frac{c_x}{x_{\text{size}}}, \frac{c_y}{y_{\text{size}}}, L_{11}, \sqrt{2} L_{12}, L_{22} \right), \qquad L = \log \widehat{S}.$$

This produces identical features for geometrically identical ellipses, including periodic angles, axis swaps and circles.

Table 3: Approximate implementation areas for the ellipse feature transformation

Module/function	**Required modification**
`shapes.py` or a geometry-feature helper	Introduce a differentiable transformation from encoded ellipse parameters to invariant shape-matrix features. Preserve raw geometric decoding.
`gaussian.py`	Train the GP using transformed features and apply the identical transformation during posterior prediction.
`sampling.py`: `rank_initial_condition_pool`	Calculate ellipse novelty using transformed geometric distances instead of raw parameter distances.
Archive/restart-diversity helpers	Use the same geometric metric for selecting distinct archived starts and detecting near-duplicates.
`optimization.py` / acquisition interface	Ensure candidate points are transformed consistently when the acquisition function queries the GP. Retain raw encoded parameters as the acquisition-optimisation variables.
Tests	Verify invariance under periodic angles, axis swapping and arbitrary circle orientation, including gradient behaviour.

The transformation must be applied consistently to both training and candidate points. Transforming only stored observations would be incorrect. Its implementation must also preserve differentiability for acquisition optimisation. Matrix-logarithm evaluation near repeated eigenvalues (circles) requires particular numerical care. For multiple holes, retain consistent canonical hole ordering and shape-type handling. If triangle and ellipse configurations share a GP, the overall feature representation must accommodate both types without introducing ambiguous dummy features. Functions that should remain physically unchanged:

- Ellipse rasterisation and FEM evaluation.
- Geometric containment and area calculations.
- SAT-based clearance and non-overlap constraints.
- The raw encoded design variables and their permitted bounds

The benefit should be tested through an ablation comparing raw features against symmetry-invariant features. Representation invariance is mathematically guaranteed; improved BO performance is not. The proposed SAT accuracy audit is a separate numerical-validation task.

7 Structured Work Packages

7.1 WP1: Reproducible Experiments and Baseline Evaluation

Addresses Questions 1-3 Objective: Establish reliable and comparable experimental conditions before changing optimiser behaviour.

- Persist complete evaluations, GP-fitting diagnostics, restart metadata, configurations and random-generator states.
- Separate random-number streams by optimisation phase and iteration.
- Introduce an option to disable adaptive restart allocation during controlled experiments.
- Preserve the current GP and restart-selection implementations as experimental baselines.
- Add standardised reporting of final compliance, best-so-far trajectories, evaluations-to-target, numerical failures and runtime.

Completion criterion: Fixed-seed runs can be reproduced, selected GP states can be replayed, and configurations can be compared under controlled conditions.

7.2 WP2: GP Stability and Deterministic-Response Modelling

Addresses: Question 1. Objective: Identify and prevent the surrogate collapse observed for seed 76.

- Audit the iteration-11 geometry and independently repeat its FEM evaluation.
- Replay GP fits immediately before and after the reported failure.
- Compare the current learned-noise model with fixed-small-noise alternatives.
- Test a predefined numerical-nugget range in consistent output units.
- Introduce GP health diagnostics and a recorded cold-refit fallback.
- Compare candidate models using the staged paired-seed experiment in Section A.

Completion criterion: Reduced frequency of degenerate GP fits without deterioration in predictive or BO performance. Priority: High. Other restart experiments are difficult to interpret while the surrogate can collapse.

7.3 WP3: Improved Global Restart Selection

Addresses: Question 2. Objective: Make acquisition/novelty weighting influence every selected global start.

- Expose validated acquisition/novelty weights in OptimizationConfig.
- Implement greedy weighted selection with novelty recalculated after each start.
- Apply consistent score normalisation and duplicate/feasibility filtering.
- Retain the existing selection strategy as an experimental comparison option.
- Compare start diversity, optimized acquisition values, duplicate endpoints, runtime and final compliance.

Completion criterion: The effect of weighting is measurable and its implementation matches the intended selection rule.

7.4 WP4: Controlled Global/Local Restart Budget Study

Addresses: Question 3, Change 1. Objective: Determine whether additional restarts or different source ratios improve acquisition optimisation.

- Construct reproducible, nested start sets for frozen-GP experiments.
- Compare total restart budgets of 5, 10, 15 and 30.
- For a fixed total of 15, compare several global/local allocations with adaptive control disabled.
- Measure feasible optimized LogEI, unique endpoints, optimisation failures and wall time.
- Validate promising settings using paired full-run seeds.

Completion criterion: Identify a restart budget and allocation supported by measured performance, rather than nominal counts. This package should initially preserve the existing archive-start behaviour.

7.5 WP5: Archive-Based Trust-Region Search

Addresses: Question 3, Change 2. Objective: Convert archive-based starts into genuinely local acquisition optimisation.

- Retain quality- and diversity-based selection of promising archive points.
- Introduce constrained, archive-centred local optimisation regions.
- Specify neighbourhood scaling, success criteria and contraction/expansion rules.
- Preserve an unrestricted global-restart pathway.
- Compare against unrestricted archive starts at the same evaluation budget.

Completion criterion: Demonstrate whether bounded local exploitation improves compliance or efficiency. Priority: Experimental. Do not introduce this simultaneously with WP4, since it changes what “local restart” means.

7.6 WP6: Symmetry-Invariant Ellipse Features

Addresses: Question 4. Objective: Remove redundant ellipse representations from GP modelling and geometric diversity calculations.

- Implement and numerically validate the dimensionless shape-matrix feature transformation.
- Apply it consistently in GP training, posterior prediction and acquisition evaluation.
- Use a compatible invariant metric for ellipse novelty and archive diversity.
- Preserve the existing raw representation for physical calculations.
- Validate exact symmetry equivalence and stable differentiability near circles.
- Compare raw-feature and invariant-feature BO across paired ellipse seeds.

Completion criterion: Equivalent ellipses have identical GP features, with no regression in physical feasibility or acquisition optimisation. This is a separate GP-model comparison and should not be combined with the initial noise-model investigation.

7.7 WP7: Ellipse Clearance Accuracy Audit

Addresses: Question 4. Objective: Establish whether the SAT approximation contributes materially to ellipse feasibility errors.

- Generate representative near-contact ellipse–ellipse and triangle–ellipse configurations.
- Compare the existing directional approximation against a substantially finer numerical reference.
- Measure errors in minimum clearance and feasible/infeasible classification.
- Modify the directional resolution or clearance calculation only if significant errors are demonstrated.

Completion criterion: Quantified clearance accuracy around the minimum-spacing boundary.

7.8 Recommended Implementation Order

Table 4: Work-package dependencies and suggested priorities

Order	**Work package**	**Purpose**
1	WP1: Reproducibility and baselines	Establish fair comparison
2	WP2: GP reliability	Resolve fundamental surrogate instability
3	WP3: Global start selection	Improve and validate restart diversity
4	WP4: Restart budget study	Establish efficient restart allocation
5	WP5: Local trust regions	Test an additional optimisation strategy
6	WP6: Ellipse invariance	Improve ellipse GP representation
Parallel	WP7: Clearance audit	Verify numerical geometry handling

WP6 can be developed independently after WP1–WP2, rather than necessarily waiting for WP5. However, final performance comparisons should change one major modelling component at a time. Overall recommendation: Prioritise reproducibility and GP stability, then optimise restart selection and allocation. Introduce true local trust regions and ellipse-invariant GP features as separately evaluated improvements. This preserves Bayesian optimisation and avoids confusing changes in surrogate behaviour with changes in acquisition optimisation.